

System Architecture Structure: Shop

1. Client Layer (Frontend)

- React + Vite SPA
- Routing and page modules
- State management and persistence
- API client integration
- Session sync and idle timeout

2. API Layer (Backend)

- Hono application
- Entry points (Vercel/Render)
- Middleware (CORS, auth, admin guard)
- Route modules (auth, products, orders, admin, settings, contact)

3. Data Layer

- Prisma ORM
- PostgreSQL database
- Core entities (User, Product, Order, OrderItem, Settings)

4. External Integrations

- WhatsApp link generation and handoff
- Hosting platforms
- Environment and runtime configuration

5. Core Runtime Flows

1.1 Authentication Flow

1.2 Product Browse Flow

1.3 Order + Checkout Flow

1.4 Contact/Complaint via WhatsApp Flow

1.5 Admin Operations Flow

6. Security and Session Controls

- JWT issuance and verification
- Protected route enforcement
- Session revalidation after rehydration
- Idle logout policy

7. Deployment Topology

- Local development topology
- Vercel topology
- Render topology

8. Observability and Error Handling

- Client-side error surface
- API error responses
- Logging points

9. Future Extension Points

- Payment provider integration
- Email/SMS notifications
- Caching layer
- Background jobs